

```
//Custom Exceptions
```

```
using System;
```

```
public class InsufficientBalanceException : Exception
```

```
{
```

```
    public InsufficientBalanceException(string message) : base(message) { }
```

```
}
```

```
public class InvalidAmountException : Exception
```

```
{
```

```
    public InvalidAmountException(string message) : base(message) { }
```

```
}
```

```
//BankAccount Class
```

```
using System;
```

```
public class BankAccount
```

```
{
```

```
    public string AccountHolderName { get; set; }
```

```
    public double Balance { get; private set; }
```

```
    private const double MIN_BALANCE = 1000;
```

```
    public BankAccount(string name, double initialBalance)
```

```
    {
```

```
        AccountHolderName = name;
```

```
        if (initialBalance < MIN_BALANCE)
```

```
    throw new ArgumentException("Initial balance must be at least ₹1000");
```

```
    Balance = initialBalance;  
}
```

```
public void Deposit(double amount)
```

```
{  
    if (amount <= 0)  
        throw new InvalidAmountException("Deposit amount must be greater than 0");
```

```
    Balance += amount;  
    Console.WriteLine($"₹{amount} deposited successfully.");  
}
```

```
public void Withdraw(double amount)
```

```
{  
    if (amount <= 0)  
        throw new InvalidAmountException("Withdrawal amount must be greater than 0");
```

```
    if (amount > Balance)  
        throw new InsufficientBalanceException("Insufficient balance for withdrawal");
```

```
    if ((Balance - amount) < MIN_BALANCE)  
        throw new InsufficientBalanceException("Cannot go below minimum balance ₹1000");
```

```
    Balance -= amount;  
    Console.WriteLine($"₹{amount} withdrawn successfully.");  
}
```

```
public void CheckBalance()
```

```
{
```

```

        Console.WriteLine($"Current Balance: ₹{Balance}");
    }
}

//Main Program

using System;

class Program
{
    static void Main()
    {
        try
        {
            BankAccount account = new BankAccount("John", 5000);

            // Valid operations
            account.Deposit(2000);
            account.Withdraw(3000);
            account.CheckBalance();

            // Invalid scenarios
            Console.WriteLine("\n--- Testing Invalid Cases ---");

            account.Deposit(-500);    // InvalidAmountException
            account.Withdraw(10000);  // InsufficientBalanceException
            account.Withdraw(4500);   // Minimum balance violation
        }
        catch (InvalidAmountException ex)
        {
            Console.WriteLine("Invalid Amount Error: " + ex.Message);
        }
    }
}

```

```
}  
catch (InsufficientBalanceException ex)  
{  
    Console.WriteLine("Balance Error: " + ex.Message);  
}  
catch (ArgumentException ex)  
{  
    Console.WriteLine("Argument Error: " + ex.Message);  
}  
catch (Exception ex)  
{  
    Console.WriteLine("Unexpected Error: " + ex.Message);  
}  
finally  
{  
    Console.WriteLine("\nTransaction process completed.");  
}  
}  
  
//
```